THE USE CASE

Turn hidden headroom into capacity the owner can safely sell

We help a data-centre owner safely sell capacity they already have but cannot confidently offer to a customer today.

FIND
Where is capacity trapped?
Allocated • reserved • constrained

PROVE
Is it usable in real conditions?
Peak • weather • maintenance

RELEASE
How much can be committed?
Location • conditions • approval

ASSURE
What would invalidate it?
Change • alert • recalculation

Find the hidden capacity. **Prove it is usable.** **Release it safely.** **Keep proving it.**

For operations, commercial, capacity and facilities-engineering teams deciding how much load can be contracted, where and under which conditions.

A REAL CAPACITY SITUATION

The hall appears to have 1.9 MW free—but most of it cannot be sold as-is

8 MW OPERATING HALL

7.2 MW is allocated; only 5.3 MW appears at coincident peak.

+700 kW

An existing customer may still ramp

PDU

One local path is phase-imbalanced

COOLING

Design weather + one chiller unavailable binds

MAINTENANCE

UPS work begins next month

600 kW

Delayed customer still holds a reservation

WHAT CAN ACTUALLY BE DONE

Release a safe increment now, unlock more through targeted work, renegotiate reserved-idle capacity—and reject unsupported headroom.

THE DECISION

400 kW now • 500 kW after work • 600 kW to renegotiate

A commercial commitment must survive contractual ramps, local constraints, maintenance, design weather and agreed resilience states.

WHAT THE PRODUCT PROVIDES

A location-specific capacity release decision—not another utilisation dashboard

1.1 MW RECOVERABLE

Hall A can release capacity in three controlled steps.

Release now	400 kW supported by current evidence
Release after action	300 kW after phase rebalancing
Commercial recovery	400 kW held by an expired reservation
Governing condition	Cooling binds above 6.4 MW in design weather with Chiller 3 unavailable
Validity window	Valid until UPS maintenance begins on 14 September

CAPACITY STATUS

Now	Evidence supports commitment
With action	A named change unlocks it
Commercial	Held by reservation or contract
Conditional	Valid within explicit limits
Not releasable	Hard engineering or contract limit
Recalculate	A material change invalidates it

Engineers approve technical release. Authorised commercial owners approve the customer commitment; the system keeps the evidence and validity conditions visible.

WHY THIS IS USEFUL

Release the cheapest safe increment first—and keep every MW defensible

TODAY

Installed, allocated, measured and safely releasable capacity are treated as competing numbers.

- Current under-use looks sellable
- Site totals hide local constraints
- Normal weather becomes the planning case
- Recovery options compete informally
- A study becomes stale after approval

WITH CAPACITY RECOVERY

Commercial inventory is tied to location, constraints, scenarios and engineering approval.

- ✓ **Respect customer ramps, reservations and future demand**
- ✓ **Find the breaker, PDU, phase, UPS or cooling zone that binds**
- ✓ **Test design weather, maintenance and approved contingencies**
- ✓ **Rank MW unlocked by cost, time, operational risk and approval need**
- ✓ **Recalculate when contracts, topology, equipment or conditions change**

Practical value: more sellable MW, faster customer placement, deferred capex, recovered reservations and protected resilience.

WHAT MAKES IT POSSIBLE

Join engineering limits, live demand and contractual rights in one capacity model

THE DATA WE NEED

1	**Physical paths**	Electrical + cooling topology • ratings • settings
2	**Actual behaviour**	EPMS/BMS/DCIM • branch and zone telemetry
3	**Asset states**	Maintenance • degradation • availability
4	**Customer rights**	Allocations • ramps • reservations • expiry
5	**Future demand**	Sales pipeline • deployment • density profile
6	**Assurance cases**	Peak • weather • maintenance • contingency

HOW THE ANSWER IS BUILT

1. **Reconcile capacity truth**
   Join installed, allocated, reserved and measured capacity
2. **Find the binding constraint**
   Calculate power and cooling limits at every relevant node
3. **Stress the proposed release**
   Test ramp, peak, design weather, maintenance and failure states
4. **Recommend and keep revalidating**
   Release, recover or reject; monitor every invalidation trigger

ML forecasts demand inside the envelope. Deterministic engineering rules define the safe limit. Governed workflow routes the approval.

What must be true before we test the use case

Start with one constrained hall and one real proposed deployment—not a site-wide theoretical capacity study.

✓	**Validated topology**	Engineers can confirm electrical and cooling paths to proposed racks
✓	**Granular telemetry**	Sufficient history below site level, with known quality and states
✓	**Commercial truth**	Allocations, ramp rights, reservations and a real buyer or deployment
✓	**Scenario definition**	Maintenance, weather, peak and contingency states are agreed
✓	**Named decision owners**	Electrical, mechanical and commercial reviewers can act

If topology, cooling, contractual rights or engineering review are missing, position this as capacity-evidence readiness—not capacity release.

Plain language on the left; technical capability on the right

WHAT THE CLIENT HEARS	WHAT WE BUILD	WHAT IT DOES
Rebuild how power and cooling reach the rack	**Topology reconstruction**	Convert drawings, registers and monitoring points into a dependency graph
Join meters, assets, customers and contracts	**Entity resolution**	Connect physical assets, telemetry, allocations and rights
Calculate the safe engineering limit	**Deterministic constraint model**	Apply ratings, derating, redundancy and cooling delivery at each node
Forecast future demand and coincidence	**Hierarchical forecasting**	Estimate customer, circuit, zone and site demand with uncertainty
Test every important operating condition	**Scenario engine**	Simulate peak, weather, maintenance, degradation and failure
Find the best placement or recovery action	**Constrained optimisation**	Rank feasible locations and actions by MW, cost, time and risk

Control boundary: no autonomous setpoint, protection, topology, cooling, allocation or contract change—and no technical or commercial approval.